

## 05 Big Data Life Cycle - KirkYagami

### Big Data Analytics Life Cycle

#### I. Introduction

The Big Data Analytics Life Cycle is a comprehensive process for handling and analyzing large, complex datasets. It consists of nine phases, each crucial for turning raw data into actionable insights.

#### II. The Nine Phases

##### 1. Business Case/Problem Definition

- **Objective:** Understand the business context and define the problem.
- **Key Activities:**
  - Learn about the business domain
  - Identify specific problems or opportunities
  - Frame the business problem as an analytics challenge
  - Estimate potential gains and required resources
  - Determine if it's truly a big data problem (volume, velocity, variety)
- **Importance:** Sets the direction for the entire project

##### 2. Data Identification

- **Objective:** Locate appropriate datasets for analysis.
- **Key Activities:**
  - Research similar cases in other companies
  - Identify internal data sources (e.g., feedback forms, existing software)
  - Explore external data sources (e.g., third-party providers)
- **Consideration:** Data relevance to the business case is crucial

##### 3. Data Acquisition and Filtration

- **Objective:** Gather and initially clean the data.
- **Key Activities:**
  - Collect data from identified sources
  - Remove corrupt or irrelevant data
  - Store a compressed copy of filtered data for potential future use
- **Challenge:** Dealing with mostly unstructured data

##### 4. Data Extraction

- **Objective:** Ensure all data is compatible with the analysis scope.
- **Key Activities:**
  - Identify incompatible data entries
  - Transform incompatible data to fit the analysis requirements
- **Importance:** Prepares data for more detailed cleaning and validation

##### 5. Data Munging (Validation and Cleaning)

- **Objective:** Thoroughly clean and validate the data.
- **Key Activities:**
  - Remove invalid data
  - Establish and apply complex validation rules
  - Handle null entries (e.g., fill from similar datasets or remove)
- **Importance:** Ensures data quality for accurate analysis

##### 6. Data Aggregation & Representation (Storage)

- **Objective:** Combine datasets and prepare for storage.
- **Key Activities:**
  - Join multiple datasets using common fields
  - Consider automation for large-scale operations
- **Challenge:** Handling potentially very large amounts of data

##### 7. Exploratory Data Analysis

- **Objective:** Analyze the data to extract insights.
- **Types of Analysis:**
  1. Confirmatory Analysis:
    - Test pre-existing hypotheses

- Provide definitive answers to specific questions
2. Exploratory Analysis:
- Discover patterns and relationships in the data
  - Answer "why" a phenomenon occurred

- **Importance:** Core step where insights are generated

## 8. Data Visualization (Preparation for Modeling and Assessment)

- **Objective:** Represent findings visually for easy interpretation.
- **Key Activities:**
  - Use visualization tools to create graphics
  - Ensure visualizations are understandable to business users
- **Benefits:**
  - Aids in result interpretation
  - Can reveal answers to unasked questions

## 9. Utilization of Analysis Results

- **Objective:** Apply insights to business decisions.
- **Key Activities:**
  - Make data-driven decisions
  - Optimize and refine business processes
  - Use results as input for enhancing system performance
- **Importance:** Translates analysis into tangible business value

## III. Iterative Nature of the Process

- Phases 7 and 8 (Analysis and Visualization) may be repeated for better results
- Emphasis on error correction and refinement
- Allows for moving back from Phase 8 to Phase 7 if needed

## IV. Key Considerations

1. **Data Quality:** Crucial throughout the process, especially in phases 3-5
2. **Scalability:** Must handle large volumes of data efficiently
3. **Expertise:** Requires a mix of business knowledge and technical skills
4. **Technology:** Appropriate tools needed for each phase, especially for analysis and visualization
5. **Privacy and Ethics:** Important when dealing with sensitive data

## V. Conclusion

The Big Data Analytics Life Cycle is a comprehensive approach to extracting value from complex datasets. It requires careful planning, rigorous data handling, and skilled analysis to turn raw data into actionable business insights.



